

Financial Econometrics: Unit root test

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Impulse-Response.

Q11) AR(1)

$$y_t = \phi y_{t-1} + e_t$$

$$\mu = 0$$

"ER"

$$\lim_{j \rightarrow \infty} \rho_j = 0$$

Stationary.

Stability condition

$$\frac{\partial y_{t+j}}{\partial e_t} = \phi^j$$

IR

$$\phi = 1$$

$$|\phi| < 1$$

$\phi > 1$: $\bar{y}$ th permanent
 $\phi = 1$: " 0.2522
 $\phi < 1$: transitory

AR(2)

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + e_t$$

(1st step) (state-space form) $\Rightarrow$ 1st order model

$$\begin{bmatrix} y_t \\ y_{t-1} \end{bmatrix} = \begin{bmatrix} \phi_1 & \phi_2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} y_{t-1} \\ y_{t-2} \end{bmatrix} + \begin{bmatrix} e_t \\ 0 \end{bmatrix}$$

$\tilde{z}_t$ (y_{t-1} = y_{t-1}) $T \cdot \tilde{z}_t + e_t^2$

2nd Step)

(bold form)

$$\tilde{z}_t = F \cdot \tilde{z}_{t-1} + \tilde{e}_t$$

$$\rightarrow F \tilde{z}_{t-2} + e_{t-1} \Rightarrow F \cdot F^2(\tilde{z}_{t-3}) + F \cdot \tilde{e}_{t-2} + \tilde{e}_{t-1}$$

$$\tilde{z}_t = \tilde{e}_t + F \cdot \tilde{e}_{t-1} + F^2 \tilde{e}_{t-2} + F^3 \tilde{e}_{t-3} + F^4 \tilde{e}_{t-4} + \dots$$

Stationary

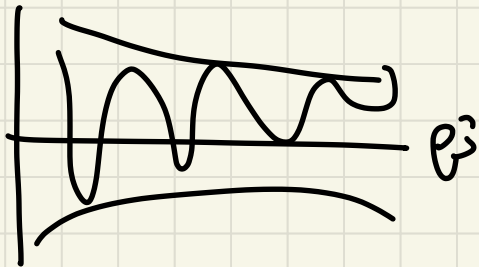
$$\frac{\partial Y_{t+j}}{\partial e_t} = \frac{\partial Y_t}{\partial e_{t-j}} \rightarrow ? \quad (F^j)$$

$$\begin{bmatrix} Y_t \\ Y_{t-1} \end{bmatrix} = \dots \begin{bmatrix} f_{1j}^j & f_{2j}^j \\ f_{2j}^j & f_{1j}^j \end{bmatrix} \begin{bmatrix} e_{t-j} \\ e_{t-j-1} \end{bmatrix} + \dots$$

$$F^j = \begin{bmatrix} \phi_1 & \phi_2 \\ 1 & 0 \end{bmatrix}^j \Rightarrow I-R \text{ 구조 !!}$$

III

$$\begin{bmatrix} Y_t - Y_{t-1} \\ Y_t - \Delta \mu \end{bmatrix} = \dots \begin{bmatrix} e_{t-1} \\ e_{t-2} \end{bmatrix}$$



$Fx = \lambda x$ 이 방정식을 풀다.
 $n \times n$ matrix, $\lambda \neq 0$

$$Fx - \lambda x = 0$$

$$(F - \lambda I_n) x = 0$$

$$\det(F - \lambda I_n) = 0 \text{ 일 때 } \lambda \neq 0 \text{ for } \lambda$$

$$\left| \begin{pmatrix} \phi_1 & \phi_2 \\ 1 & 0 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} \right| = 0$$

$$\left| \begin{pmatrix} \phi_1 - \lambda & \phi_2 \\ 1 & -\lambda \end{pmatrix} \right| = 0$$

$$\lambda^2 - \phi_1 \lambda - \phi_2 = 0 \quad \text{특성 방정식}$$

$$\Rightarrow \lambda_1, \lambda_2 \quad \text{특성근}$$

i) When $\lambda = \lambda_1$, When $\lambda = \lambda_2$

$$F x_1 = \lambda x_1, \quad F x_2 = \lambda x_2$$

$$\underline{x_1' x_1 = 1}$$

$$\underline{x_2' x_2 = 1}$$

III) λ on the real axis.

$\lambda_1, \lambda_2, \dots, \lambda_n$ are the eigenvalues.

$$F x_1 = \lambda x_1$$

$$F x_2 = \lambda x_2$$

$\mathbb{R}^2$ vector

$$F \underbrace{\begin{bmatrix} x_1 & x_2 \end{bmatrix}}_C = \underbrace{\begin{bmatrix} x_1 & x_2 \end{bmatrix}}_C \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$$

$$F \cdot C = C \cdot \Lambda$$

$\mathbb{R}^2$ vector
 $\approx$ basis

$\mathbb{R}^2$ basis
 orthonormal

$$F = C \cdot \Lambda \cdot C^{-1}$$

$$\dots \rightarrow F^3 = (C \Lambda C^{-1}) (C \Lambda C^{-1}) (C \Lambda C^{-1}) \dots = (C \Lambda C^{-1}) : \boxed{C \Lambda^3 C^{-1}}$$

$$\frac{\partial Y(t, \bar{s})}{\partial e_t} = \frac{\partial Y_t}{\partial e_{t-j}} = ?$$

(1.1) element of $(F^{\bar{s}}) \Rightarrow \underbrace{\omega_1 \lambda_1^{\bar{s}} + \omega_2 \lambda_2^{\bar{s}}}$

$\Rightarrow$ Stability condition?

$$\text{if } |\lambda_1| < 1, \quad |\lambda_2| < 1, \quad \lim_{\bar{s} \rightarrow \infty} Q^{\bar{s}} = 0$$

11

가짜 회귀
→
Spurious regression problem

General expression for a time series variable

Trend-stationary variable

Difference-stationary variable

Lag operator

Unit root tests

Outline

Spurious regression problem

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Spurious regression problem



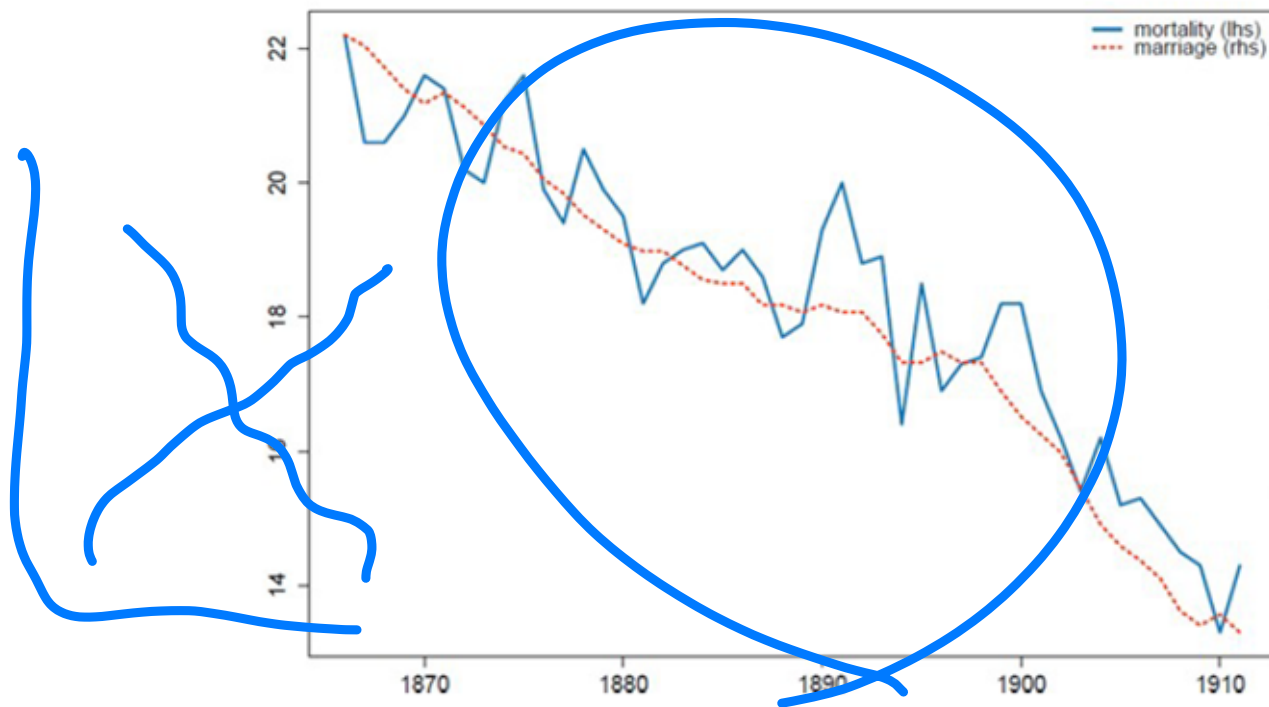
Figure: George Udny Yule

Yule, G. U. (1926). Why Do We Sometimes Get Nonsense-
~~Correlations between~~ Time Series? A Study in Sampling and the
Nature of Time Series. Journal of Royal Statistical Society, 89, 1-
64

Spurious regression problem

748 621

non-stationary



⇒ spurious regression.

DR1) 74.21% & 12.02%

$$C_t = \beta_1 + \beta_2 S_t + e_t$$



Figure: Mortality rate and Marriage

Early studies that consider the effects of incorporating variables that contain a trend in a regression model include the work of Yule (1926), which considers the possible relationship between mortality and marriage.

Spurious regression problem

Figure 1: Mortality and Marriage

Dependent Variable: Mortality

	Coefficient	Std. Error	t-value	prob
constant	-13.88	1.57	-8.82	0
marriage	<u>0.04</u>	0	<u>20.51</u>	0
R^2	0.91			

Table 1: Regression results for mortality and marriage - Levels

Dependent Variable: Δ Mortality

	Coefficient	Std. Error	t-value	prob
constant	-0.133	0.21	-0.63	0.531
Δ marriage	<u>0.011</u>	0.043	<u>0.27</u>	0.788
R^2	0.001			

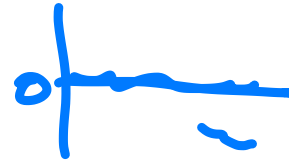
Table 2: Regression results for mortality and marriage - First difference

Spurious regression problem

- ▶ The two variables look highly related in the regression result that uses the levels of the two variables although they should not be in reality. This problem is referred to as the spurious regression problem.
- ▶ After taking the first difference and using the differenced data in the regression model, the result looks much more sensible.
- ▶ What is happening here?

Outline

Spurious regression problem



General expression for a time series variable

Trend-stationary variable **TSP**

Difference-stationary variable **DSP**

Δ

Lag operator

Unit root tests

General expression for a time series variable

$$Y_t = \text{non-stationary component} + \text{stationary component}$$

Handwritten notes: "Deterministic" with an arrow pointing to the non-stationary component; "stochastic" with an arrow pointing to the stationary component; "ARIMA" with an arrow pointing to the entire equation.

- In the above expression, we ignore any seasonal component.

- In the above expression of Y_t , the non-stationary component could be deterministic (i.e., without any randomness) or stochastic (i.e., with randomness).

- What is the most important condition for stationarity? Simply speaking, stationarity means that similar data patterns repeat over and over again. How do we measure similar patterns for a time series variable? We can use mean (expected value) and variance (amount of uncertainty). For example, if $E[y_t]$ and $Var[y_t]$ are constant over time, we can say y_t is stationary. If, the mean and variance of y_t vary over time, y_t is non-stationary.

Outline

1982

real GDP

$$Y_t = \phi Y_{t-1} + e_t$$

Spurious regression problem

→ If $\phi = 1$, Y_t : permanent
($\frac{1}{2} \frac{1}{2} \frac{1}{2} \frac{1}{2}$)

General expression for a time series variable

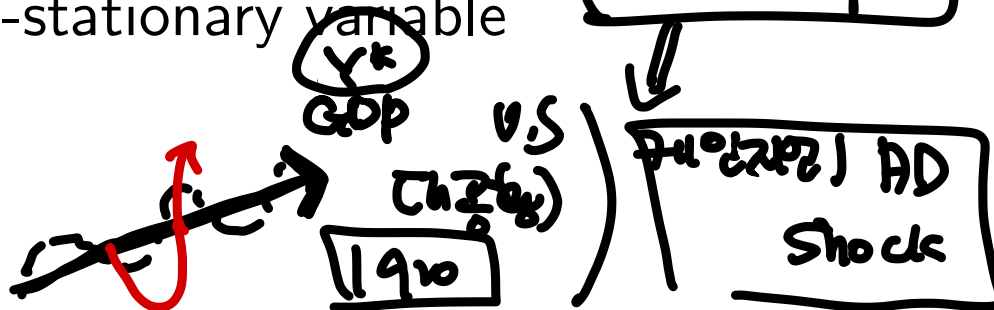
Trend-stationary variable

Difference-stationary variable

If $\phi < 1$, Y_t : transitory
(AD.)

real GDP

Lag operator



Unit root tests

⇒ $Y \uparrow$
• oil shock (stagflation)
 $\pi \uparrow$ $Y \downarrow$

→ $\frac{1}{2} \frac{1}{2} \frac{1}{2} \frac{1}{2}$
AD.
PS
AD.
PS

Deterministic trend and trend-stationary variable $(D.S.P)$

Consider a time series variable that follows:

$$Y_t = \underbrace{\alpha_0 + \alpha_1 t}_{\text{Non-stationary}} + \underbrace{u_t}_{\text{Stationary}}, \quad u_t \sim \text{i.i.d.} N(0, \sigma^2).$$

where α_0 is the intercept of the time trend, α_1 is the slope of the time trend, t is the time index, and u_t is the stationary component.

- Note that $\alpha_0 + \alpha_1 t$ is the deterministic time trend of Y_t and it is the non-stationary component at the same time.
- The stationary component of Y_t is the white noise error u_t .

Deterministic trend and trend-stationary variable

A more general stationary component is a serially correlated series such as

$$\text{AR}(p): \underline{u_t} = \phi_1 u_{t-1} + \phi_2 u_{t-2} + \dots + \phi_p u_{t-p} + e_t,$$

and

$$\text{MA}(q): \underline{u_t} = e_t + \theta_1 e_{t-1} + \theta_2 e_{t-2} + \dots + \theta_q e_{t-q},$$

and

$$\boxed{\text{ARMA}(p,q)}: u_t = \phi_1 u_{t-1} + \dots + \phi_p u_{t-p} + e_t + \theta_1 e_{t-1} + \dots + \theta_q e_{t-q},$$

where $e_t \sim i.i.dN(0, \sigma^2)$ and $\sum_{\ell=1}^p \phi_\ell < 1$.

Stationary

Deterministic trend and trend-stationary variable

If u_t is a stationary process, it has constant mean and variance. In such a case, what is the expected value of y_t ? By taking the (unconditional) expectation to Y_t , we can easily see that

$$E[Y_t] = \alpha_0 + \alpha_1 t + \mu_u$$

where $\mu_u = E[u_t] \neq 0$ is the time-invariant mean of u_t .

- Because $E[Y_t]$ changes with t , Y_t is non-stationary.

$u_t \sim iid N(0, \sigma_u^2)$

Deterministic trend and trend-stationary variable

- We can easily extract the stationary component from Y_t by eliminating the trend component.

Suppose that the true process of Y_t is given by

$$\rightarrow Y_t = \alpha_0 + \alpha_1 t + u_t \quad (\text{pop. eqn.})$$

The regression model is given by

$$\text{OLS)} \quad Y_t = \alpha_0 + \alpha_1 t + \epsilon_t, \quad \epsilon_t \sim \text{i.i.d. } N(0, \sigma^2) \quad \checkmark$$

where α_0 is the intercept, α_1 is the slope of the time trend, ϵ_t is the error term.

$$Y_t - \hat{\alpha}_0 - \hat{\alpha}_1 \cdot t$$

- By running OLS, we can get the OLS residual $\hat{\epsilon}_t = Y_t - \hat{\alpha}_0 - \hat{\alpha}_1 t$ which represents the short-run deviation of Y_t from its long-run trend.

Deterministic trend and trend-stationary variable

- ▶ Note that there is no endogeneity issue in the above regression model because the error term ϵ_t is not correlated with the time trend. (Think about why!)
- ▶ Thus, we can correctly estimate the time trend. In other words, we can correctly estimate α_0 and α_1 .
- ▶ **The OLS residual $\hat{\epsilon}_t$ approximates the stationary component of y_t , which is u_t .**
- ▶ **Because Y_t follows a stationary process after removing the deterministic trend, it is called a 'trend-stationary variable'.**

Outline

Spurious regression problem

General expression for a time series variable

Trend-stationary variable

Difference-stationary variable

Lag operator

Unit root tests

Random walk process

The simplest model of a variable with a stochastic trend is the random walk, which depends on past values of itself and Gaussian white noise errors,

$$Y_t = Y_{t-1} + u_t, \quad u_t \sim \text{i.i.d. N}(0, \sigma^2)$$

For the random walk model, we can easily derive its alternative form as

$$\begin{aligned} Y_t &= Y_{t-1} + u_t \\ &= Y_{t-2} + u_{t-1} + u_t \\ &= Y_{t-3} + u_{t-2} + u_{t-1} + u_t \\ &\vdots \\ &= Y_0 + \sum_{j=0}^{t-1} u_{t-j} \end{aligned}$$

Therefore each shock, u_t will influence subsequent values of Y_t . This would imply that a shock to a random walk has a permanent effect on the time series variable.

Random walk process

If we assume that Y_0 is equal to zero for simplicity, without any loss of generality, we can write the random walk as,

$$Y_t = \sum_{j=0}^{t-1} u_{t-j}$$

- ▶ **The time series variable Y_t which follows a random walk process is a non-stationary variable.**
- ▶ Why? We can easily show from the above equation that

$$E[Y_t] = 0$$

$$\text{Var}[Y_t] = \text{Var}[u_1] + \dots + \text{Var}[u_t] = \sigma^2 + \sigma^2 + \dots + \sigma^2 = t\sigma^2.$$

- ▶ **The variance of y_t depends on t . Because the variance increases with t , we can say y_t is a non-stationary variable.**

Random walk process

- ▶ For the random walk process, the non-stationary component is $\sum_{j=0}^{t-1} u_{t-j}$ and there is no stationary component.
- ▶ The non-stationary component can be considered as a stochastic trend component.
- ▶ How do we transform the non-stationary variable into a stationary variable? It is simple. The first difference of Y_t can be shown to follow a white noise or an error term

$$\Delta Y_t = u_t, \quad u_t \sim \text{i.i.d. N}(0, \sigma^2)$$

where $\Delta Y_t = Y_t - Y_{t-1}$.

- ▶ Because the mean (0) and variance (σ^2) of $\Delta Y_t = u_t$ is constant over time, ΔY_t is a stationary variable.

Random walk process

- ▶ In this case, Y_t is referred to as a **difference-stationary variable** because after taking difference, it becomes a **stationary variable**.
- ▶ If a variable, y_t could be made stationary after differencing it once, it is integrated of the first order.
- ▶ We use the notation $I(1)$ to describe such a process.
- ▶ Stationary random variables, such as ΔY_t are thus integrated of order zero (i.e. $\Delta Y_t \sim I(0)$).
- ▶ If it is necessary to take the second difference to achieve stationarity, where $\Delta^2 Y_t \sim I(0)$, such a process is integrated of the second order, where we would use the notation, $I(2)$.

Random walk process with a drift

Adding a constant term to the random walk model results in a random walk with drift, which may be expressed as,

$$Y_t = \alpha + Y_{t-1} + u_t, \quad u_t \sim \text{i.i.d.} N(0, \sigma^2)$$

Using recursive substitution we can show that the random walk with drift can be written as a function of a deterministic trend and stochastic term,

$$\begin{aligned} Y_t &= \alpha + Y_{t-1} + u_t \\ &= 2\alpha + Y_{t-2} + u_{t-1} + u_t \\ &= 3\alpha + Y_{t-3} + u_{t-2} + u_{t-1} + u_t \\ &\vdots \\ &= t\alpha + Y_0 + \sum_{j=0}^{t-1} u_{t-j} \end{aligned}$$

Random walk process with a drift

Let's assume $y_0 = 0$ once again for simplicity. Then, the above equation is simplified to

$$Y_t = t\alpha + \sum_{j=0}^{t-1} u_{t-j}$$

- In contrast with the random walk model without drift, a random walk with drift now also contains a deterministic trend, which results from the inclusion of the parameter, α that influences the slope of the deterministic trend.

Random walk process with a drift

Let's assume $y_0 = 0$ once again for simplicity. Then, the above equation is simplified to

$$Y_t = t\alpha + \sum_{j=0}^{t-1} u_{t-j}$$

- ▶ However, in contrast with the trend stationary model, the deviations from the deterministic trend are not stationary. This would imply that each u_{t-j} will permanently influence the value of Y_t even after removing the deterministic trend from the series.
- ▶ The non-stationary component of Y_t is given by the sum of the deterministic time trend, $t\alpha$, and the stochastic time trend $\sum_{j=0}^{t-1} u_{t-j}$. And there is no stationary component.

More general non-stationary process

We can consider a more general time series process than the previous sections. Consider the sum of two processes as

$$Y_t = x_t + z_t$$

where x_t is the non-stationary component and z_t is the stationary component.

- ▶ The resultant process Y_t will be a non-stationary variable.
- ▶ The non-stationary component can be a random walk process with drift or without drift.
- ▶ The stationary component can be a stationary AR process, MA process, or ARMA process.

Outline

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Lag operator

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Lag operator

L : Lag operator

- It is a mathematical operator that makes it very easy to handle a time series variable.

$$LY_t = Y_{t-1}$$

$$L^2 Y_t = Y_{t-2}$$

$$L^j Y_t = Y_{t-j}$$

- A reverse order is also possible.

$$L^{-1} Y_t = Y_{t+1}$$

$$L^{-j} Y_t = Y_{t+j}$$

Lag operator

Consider AR(2) model

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma^2)$$

- ▶ With the lag operator, we can rewrite the AR(2) model as

$$\begin{aligned} Y_t &= \phi_1 L Y_t + \phi_2 L^2 Y_t + \varepsilon_t, \\ (1 - \phi_1 L - \phi_2 L^2) Y_t &= \varepsilon_t, \end{aligned}$$

- ▶ Or we can use $\phi(L)$ to denote the AR polynomial

$$\phi(L) Y_t = \varepsilon_t,$$

where $\phi(L) = (1 - \phi_1 L - \phi_2 L^2)$.

Lag operator

Consider ARMA(2,1) model

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \varepsilon_t + \theta_1 \varepsilon_{t-1}, \quad \varepsilon_t \sim N(0, \sigma^2)$$

► With the lag operator,

$$\phi(L) Y_t = \theta(L) \varepsilon_t,$$

where $\phi(L) = (1 - \phi_1 L - \phi_2 L^2)$ and $\theta(L) = (1 + \theta_1 L)$.

Lag operator

An ARMA(p,q) model can be written as

$$\phi(L)Y_t = \theta(L)\varepsilon_t,$$

where $\phi(L) = (1 - \phi_1 L - \phi_2 L^2 - \dots - \phi_p L^p)$ and $\theta(L) = (1 + \theta_1 L + \dots + \theta_q L^q)$.

Lag operator

- ▶ The inverse of the first order lag polynomial is given by

$$(1 - \phi L)^{-1} = 1 + \phi L + \phi^2 L^2 + \dots + \phi^j L^j + \dots$$

when $|\phi| < 1$.

- ▶ $(1 - \phi L)^{-1}(1 - \phi L) = 1$
- ▶ If $\phi(L) = (1 - \phi_1 L - \phi_2 L^2 - \dots - \phi_p L^p)$, then $\phi(L = 1) = \phi(1) = (1 - \phi_1 - \phi_2 - \dots - \phi_p)$.

Lag operator and eigen-value

Example of $AR(2)$: $Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \varepsilon_t$ In order to check stationarity, we use eigen-values...

$$\begin{aligned}\det(\mathbf{F} - \lambda \mathbf{I}_2) &= \det \left(\begin{pmatrix} \phi_1 & \phi_2 \\ 1 & 0 \end{pmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right) \\ &= \det \begin{pmatrix} \phi_1 - \lambda & \phi_2 \\ 1 & -\lambda \end{pmatrix} \\ &= \lambda^2 - \phi_1 \lambda - \phi_2\end{aligned}$$

The eigenvalues of $\mathbf{F}$ solve the characteristic equation

$$\lambda^2 - \phi_1 \lambda - \phi_2 = 0.$$

If two solutions meets $|\lambda_1| < 1, |\lambda_2| < 1$, Y_t is a stationary process.

Lag operator

- ▶ The Condition for the eigen-values look similar to the AR polynomial. (by treating L as a unknown variable)
- ▶ Consider

$$\lambda^2 - \phi_1 \lambda - \phi_2 = 0,$$

$$1 - \phi_1 L - \phi_2 L^2 = 0$$

- ▶ There exists a relation between λ and L .

$$1 - \phi_1 \frac{1}{\lambda} - \phi_2 \frac{1}{\lambda^2} = 0,$$

$$1 - \phi_1 L - \phi_2 L^2 = 0$$

Lag operator

- ▶ There are two solutions for $\phi(L) = 0$. Let L_1 and L_2 denote the two solutions.
- ▶ There are two eigen-values, λ_1 and λ_2 .
- ▶ These are linked as $L_1 = \frac{1}{\lambda_1}$ and $L_2 = \frac{1}{\lambda_2}$.
- ▶ Thus, if $|\lambda_1| < 1$ and $|\lambda_2| < 1$, Y_t is stationary.
- ▶ Also, if $|L_1| > 1$ and $|L_2| > 1$, Y_t is stationary.

Lag operator

- ▶ An AR polynomial can be expressed as
$$\phi(L) = (1 - \phi_1 L - \phi_2 L^2 - \dots - \phi_p L^p) = (1 - \lambda_1 L)(1 - \lambda_2 L) \dots (1 - \lambda_p L).$$
- ▶ The solutions to $\phi(L) = 0$ is $L_j = \frac{1}{\lambda_j}$ for $j = 1, 2, \dots, p$.

Lag operator

- ▶ Consider $AR(2)$. Then, we have the second order polynomial like $\phi(L) = (1 - \phi_1 L - \phi_2 L^2)$.
- ▶ If $L = 1$, we have $\phi(1) = (1 - \phi_1 - \phi_2)$.
- ▶ If $L = 1$ and $\phi_1 + \phi_2 = 1$, we have $\phi(1) = (1 - \phi_1 - \phi_2) = 0$.
- ▶ This indicates that either L_1 or L_2 (or equivalently either $\frac{1}{\lambda_1}$ or $\frac{1}{\lambda_2}$) is supposed to be 1 when $\phi_1 + \phi_2 = 1$.
- ▶ Thus, when $\phi_1 + \phi_2 = 1$, there always exists a unit root and Y_t is non-stationary.
- ▶ For $AR(p)$ if $\phi_1 + \phi_2 + \dots + \phi_p = 1$, there always exists a unit root and Y_t is non-stationary.

Unit root test

- ▶ Suppose that we have a time series follows the following AR(p) process:

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \dots + \phi_p Y_{t-p} + \varepsilon_t, \quad \varepsilon_t \sim i.i.d(0, \sigma^2)$$

- ▶ If the sum of AR coefficients is one ($\phi_1 + \phi_2 + \dots + \phi_p = 1$), Y_t contains a unit root and it is a non-stationary process.
- ▶ In an unit root test, we use

$$H_0 : \phi_1 + \phi_2 + \dots + \phi_p = 1 \quad (Y_t \text{ is non-stationary})$$

$$H_a : \phi_1 + \phi_2 + \dots + \phi_p < 1 \quad (Y_t \text{ is stationary})$$

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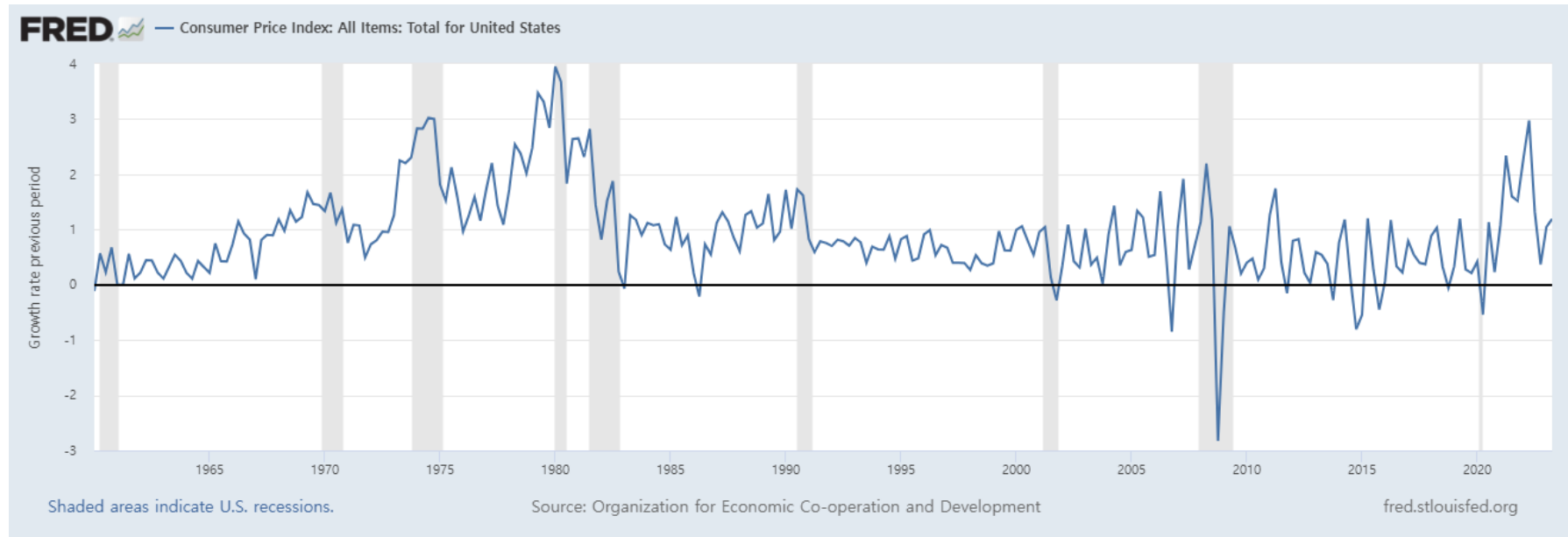


Figure: U.S. Inflation rate

- Is the U.S. inflation rate stationary or non-stationary? How can we distinguish?

Model 1: Regression with Intercept

Under the model: $Y_t = \delta + \phi Y_{t-1} + u_t$, $u_t \sim i.i.d.(0, \sigma^2)$, we have the following the null and alternative hypothesis:

$$H_0 : \delta = 0, \phi = 1$$

$$H_a : H_0 \text{ is not true.}$$

Under H_0 , the true data generating process (DGP) is given as:

$$Y_t = Y_{t-1} + u_t, \quad u_t \sim i.i.d.(0, \sigma^2)$$

Under H_a , the true data generating process (DGP) is given as:

$$Y_t = \delta + \phi Y_{t-1} + u_t, \quad u_t \sim i.i.d.(0, \sigma^2)$$

where $\phi < 1$ and $\delta \neq 0$.

Model 1: Regression with Intercept

Graphically,

Model 1: Regression with Intercept

In practice, we use a slightly modified version of the model as:

$$Y_t = \delta + \phi Y_{t-1} + u_t$$

$$Y_t - Y_{t-1} = \delta + \phi Y_{t-1} - Y_{t-1} + u_t$$

$$\Delta Y_t = \delta - (1 - \phi) Y_{t-1} + u_t$$

$$\text{Regression Model: } \Delta Y_t = \delta + \gamma Y_{t-1} + u_t$$

$$H_0 : \delta = 0, \gamma = 0 \quad (Y_t \text{ is non-stationary})$$

$$H_a : H_0 \text{ is not true. } (Y_t \text{ is stationary})$$

Motivation

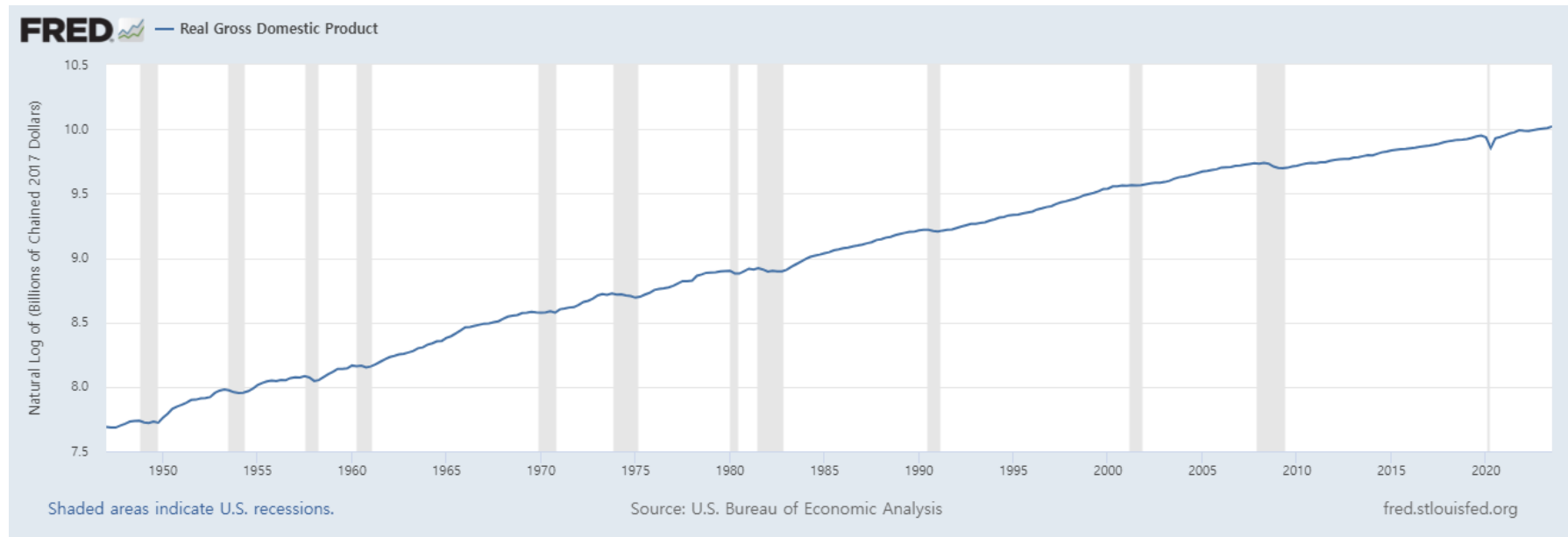


Figure: (log) U.S. real GDP

- Is the (log) U.S. real GDP stationary or non-stationary? How can we distinguish?

Model 2: Regression with time trend

Under the model: $Y_t = \delta_0 + \delta_1 t + \phi Y_{t-1} + u_t$, $u_t \sim i.i.d.(0, \sigma^2)$, we have the following the null and alternative hypothesis:

$$H_0 : \delta_1 = 0, \phi = 1$$

$$H_a : H_0 \text{ is not true.}$$

Under H_0 , the true data generating process (DGP) is given as:

$$Y_t = \delta_0 + Y_{t-1} + u_t, \quad u_t \sim i.i.d.(0, \sigma^2)$$

Under H_a , the true data generating process (DGP) is given as:

$$Y_t = \delta_0 + \delta_1 t + \phi Y_{t-1} + u_t, \quad u_t \sim i.i.d.(0, \sigma^2)$$

where $\phi < 1$ and $\delta_1 \neq 0$.

Model 2: Regression with time trend

Graphically,

Model 2: Regression with time trend

In practice, we use a slightly modified version of the model as:

$$Y_t = \delta_0 + \delta_1 t + \phi Y_{t-1} + u_t$$

$$Y_t - Y_{t-1} = \delta_0 + \delta_1 t + \phi Y_{t-1} - Y_{t-1} + u_t$$

$$\Delta Y_t = \delta_0 + \delta_1 t - (1 - \phi) Y_{t-1} + u_t$$

$$\text{Regression Model: } \Delta Y_t = \delta_0 + \delta_1 t + \gamma Y_{t-1} + u_t$$

$$H_0 : \delta_1 = 0, \gamma = 0 \quad (Y_t \text{ is non-stationary})$$

$$H_a : H_0 \text{ is not true. } (Y_t \text{ is stationary})$$

Augmented Dickey Fuller (ADF) Test

If we have

$$Y_t = \delta + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \dots + \phi_p Y_{t-p} + u_t, u_t \sim i.i.d.(0, \sigma^2)$$

or

$$Y_t = \delta_0 + \delta_1 t + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \dots + \phi_p Y_{t-p} + u_t, u_t \sim i.i.d.(0, \sigma^2)$$

how do we perform the unit root test?

That is what the ADF test is for.

Augmented Dickey Fuller (ADF) Test

AR(p) model with intercept

$$Y_t = \delta + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \dots + \phi_p Y_{t-p} + u_t, u_t \sim i.i.d.(0, \sigma^2)$$

$H_0 : \delta_1 = 0, \phi_1 + \phi_2 + \dots + \phi_p = 1$ (Y_t is non-stationary)

$H_a : H_0$ is not true. (Y_t is stationary)

Augmented Dickey Fuller (ADF) Test

AR(p) model with time trend

$$Y_t = \delta_0 + \delta_1 t + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \dots + \phi_p Y_{t-p} + u_t, u_t \sim i.i.d.(0, \sigma^2)$$

$H_0 : \delta_1 = 0, \phi_1 + \phi_2 + \dots + \phi_p = 1$ (Y_t is non-stationary)

$H_a : H_0$ is not true. (Y_t is stationary)

Models for ADF test

In practice, we use a slightly modified version of the AR models.
For example, let's say we have an AR(2) process:

$$Y_t = \delta + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + u_t$$

$$Y_t = \delta + \phi_1 Y_{t-1} + \phi_2 Y_{t-1} - \phi_2 Y_{t-1} + \phi_2 Y_{t-2} + u_t$$

$$Y_t = \delta + (\phi_1 + \phi_2) Y_{t-1} - \phi_2 (Y_{t-1} - Y_{t-2}) + u_t$$

$$Y_t - Y_{t-1} = \delta + (\phi_1 + \phi_2) Y_{t-1} - Y_{t-1} - \phi_2 (Y_{t-1} - Y_{t-2}) + u_t$$

$$Y_t - Y_{t-1} = \delta - (1 - \phi_1 - \phi_2) Y_{t-1} - \phi_2 (Y_{t-1} - Y_{t-2}) + u_t$$

$$\text{Regression Model: } \Delta Y_t = \delta + \gamma Y_{t-1} + \lambda_1 \Delta Y_{t-1} + u_t$$

where $\gamma = -(1 - \phi_1 - \phi_2)$ and $\lambda_1 = -\phi_2$.

$$H_0 : \delta = 0, \gamma = 0 \quad (Y_t \text{ is non-stationary})$$

$$H_a : H_0 \text{ is not true. } (Y_t \text{ is stationary})$$

Models for ADF test

Similarly, if we have an AR(p) process with intercept term:

Regression Model:

$$\Delta Y_t = \delta + \gamma Y_{t-1} + \lambda_1 \Delta Y_{t-1} + \lambda_2 \Delta Y_{t-2} + \dots + \lambda_{p-1} \Delta Y_{t-p+1} + u_t$$

where $\gamma = -(1 - \phi_1 - \phi_2 \dots - \phi_p)$ and $u_t \sim i.i.d(0, \sigma^2)$.

$H_0 : \delta = 0, \gamma = 0$ (Y_t is non-stationary)

$H_a : H_0$ is not true. (Y_t is stationary)

Models for ADF test

Similarly, if we have an AR(p) process with time trend:

Regression Model:

$$\Delta Y_t = \delta_0 + \delta_1 t + \gamma Y_{t-1} + \lambda_1 \Delta Y_{t-1} + \lambda_2 \Delta Y_{t-2} + \dots + \lambda_{p-1} \Delta Y_{t-p+1} + u$$

where $\gamma = -(1 - \phi_1 - \phi_2 \dots - \phi_p)$ and $u_t \sim i.i.d(0, \sigma^2)$.

$H_0 : \delta_1 = 0, \gamma = 0$ (Y_t is non-stationary)

$H_a : H_0$ is not true. (Y_t is stationary)

ADF tests in Python

The following Python code illustrates how to use the ADF test with an AR model with intercept. To include the intercept term in the model, use option **regression='c'**. The optimal order of the AR lag is selected by setting the option **autolag="AIC"**:

```
1 # import necessary packages and libraries
2 from statsmodels.tsa.stattools import adfuller
3
4 # If you would like to test for y (time series data)
   with an AR model with intercept,
5 DF_results1 = adfuller(y, autolag="AIC", regression='c'
   )
6
7 # print results
8 DF_output1 = pd.Series(DF_results1[0:4], index=[
9     "Test Statistic",
10    "p-value",
11    "#Lags Used",
12    "Number of Observations Used",])
13
14 print(DF_output1)
```

ADF tests in Python

The following Python code illustrates how to use the ADF test with an AR model with intercept. To include a linear time trend in the model, use option **regression='ct'**. The optimal order of the AR lag is selected by setting the option **autolag="AIC"**:

```
1 # import necessary packages and libraries
2 from statsmodels.tsa.stattools import adfuller
3
4 # If you would like to test for y (time series data)
   with an AR model with intercept,
5 DF_results2 = adfuller(y, autolag="AIC", regression='ct
   ')
6
7
8 # print results
9 DF_output2 = pd.Series(DF_results2[0:4], index=[
10     "Test Statistic",
11     "p-value",
12     "#Lags Used",
13     "Number of Observations Used",])
14
15 print(DF_output2)
```

ADF tests in Python

If you want to include a quadratic time trend in the model, use option **regression='ctt'**:

```
1 # import necessary packages and libraries
2 from statsmodels.tsa.stattools import adfuller
3
4 # If you would like to test for y (time series data)
  with an AR model with intercept,
5 DF_results3 = adfuller(y, autolag="AIC", regression='
  ctt')
6
7
8 # print results
9 DF_output3 = pd.Series(DF_results3[0:4], index=[
10     "Test Statistic",
11     "p-value",
12     "#Lags Used",
13     "Number of Observations Used",])
14
15 print(DF_output3)
```

Unit root tests

- ▶ **Reference for the ADF test:** Dickey, David A and Wayne A Fuller. 1979. “Distribution of the estimators for autoregressive time series with a unit root.” Journal of the American statistical association 74:427.
- ▶ There are other popular unit root tests.
- ▶ **PP test:** Phillips, Peter CB and Pierre Perron. 1988. “Testing for a unit root in time series regression.” biometrika 75:335.
- ▶ **KPSS test:** Kwiatkowski, Denis, Peter CB Phillips, Peter Schmidt, and Yongcheol Shin. 1992. “Testing the null hypothesis of stationarity against the alternative of a unit root: How sure are we that economic time series have a unit root?” Journal of econometrics 54:159.

The End.